

1. Practical: Limits and Derivatives

Date: 2026-04-14

Topic: Calculus — Change

Estimated Time: 45–60 minutes

1.1. Problem 1: Computing Limits

Evaluate the following limits. Show all steps and identify the technique used.

1.1.1. (a) Factoring

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x^2 - 5x + 6}$$

Hint: Factor both numerator and denominator.

1.1.2. (b) Rationalizing

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$$

Hint: Multiply numerator and denominator by the conjugate.

1.1.3. (c) Special Limits

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{\tan 5x}$$

Hint: Use the standard limit $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$ and express $\tan$ in terms of $\sin$ and $\cos$.

1.1.4. (d) L'Hôpital's Rule

$$\lim_{x \rightarrow 0^+} x \ln x$$

Hint: Rewrite as a quotient to get $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form.

1.1.5. (e) Limits at Infinity

$$\lim_{x \rightarrow \infty} \frac{3x^3 - 2x + 1}{5x^3 + 4x^2 - 7}$$

Hint: Divide numerator and denominator by the highest power of x .

1.2. Problem 2: Differentiation

Find the derivative of each function. Simplify your answers where possible.

1.2.1. (a) Power Rule + Chain Rule

$$f(x) = (3x^2 + 2x - 1)^5$$

1.2.2. (b) Product Rule

$$g(x) = x^2 e^{3x}$$

1.2.3. (c) Quotient Rule

$$h(x) = \frac{x^2 + 1}{x^2 - 1}$$

1.2.4. (d) Implicit Differentiation

Find $\frac{dy}{dx}$ given:

$$x^3 + y^3 = 6xy$$

1.2.5. (e) Logarithmic Differentiation

$$y = x^{\sin x}$$

1.3. Problem 3: Related Rates

A spherical balloon is being inflated such that its volume increases at a constant rate of $100\text{cm}^3/\text{s}$. Find the rate at which the radius is increasing when the radius is 5 cm.

Given:

- Volume of sphere: $V = \frac{4}{3}\pi r^3$
- $\frac{dV}{dt} = 100\text{cm}^3/\text{s}$
- Find $\frac{dr}{dt}$ when $r = 5$ cm

Visualization:

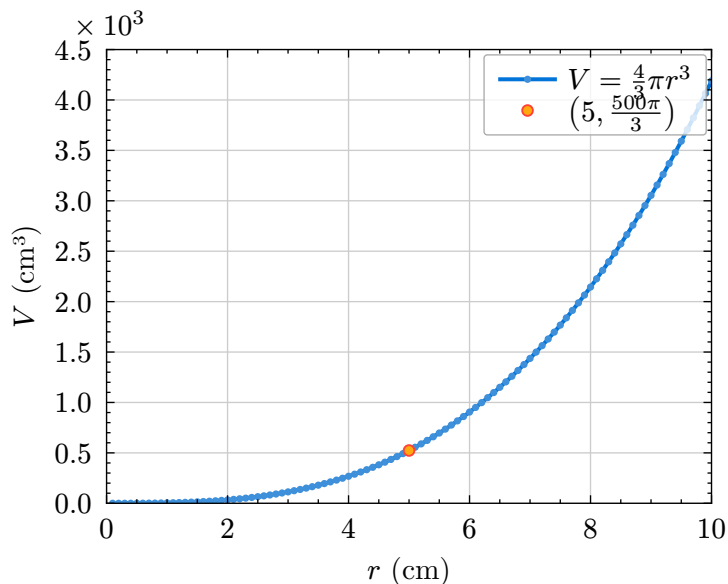


Figure 1: Volume vs. radius for a sphere. The red dot marks the point of interest.

1.4. Problem 4: Optimization

Find the dimensions of the rectangle with maximum area that can be inscribed in a semicircle of radius R .

Setup: Place the semicircle with its diameter on the x-axis, centered at the origin. The semicircle has equation $y = \sqrt{R^2 - x^2}$.

Hint: Express the area in terms of one variable, then find critical points.

1.5. Solutions (Hidden — Work Through Problems First!)

Spoiler Warning: Solutions follow on the next pages. Try each problem before checking the answer.

1.5.1. Solution 1(a): Factoring

Factor numerator: $x^2 - 9 = (x - 3)(x + 3)$

Factor denominator: $x^2 - 5x + 6 = (x - 2)(x - 3)$

$$\lim_{x \rightarrow 3} \frac{(x - 3)(x + 3)}{(x - 2)(x - 3)} = \lim_{x \rightarrow 3} \frac{x + 3}{x - 2} = \frac{6}{1} = 6$$

Technique: Factoring to resolve the removable discontinuity at $x = 3$.

1.5.2. Solution 1(b): Rationalizing

Multiply by conjugate $\sqrt{1 + x} + \sqrt{1 - x}$:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{(1 + x) - (1 - x)}{x(\sqrt{1 + x} + \sqrt{1 - x})} &= \lim_{x \rightarrow 0} \frac{2x}{x(\sqrt{1 + x} + \sqrt{1 - x})} \\ &= \lim_{x \rightarrow 0} \frac{2}{\sqrt{1 + x} + \sqrt{1 - x}} = \frac{2}{1 + 1} = 1 \end{aligned}$$

1.5.3. Solution 1(c): Special Limits

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{\tan 5x} = \lim_{x \rightarrow 0} \frac{\sin 3x}{\frac{\sin 5x}{\cos 5x}} = \lim_{x \rightarrow 0} \frac{\sin 3x \cos 5x}{\sin 5x}$$

Multiply and divide by appropriate terms:

$$= \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \cdot \frac{5x}{\sin 5x} \cdot \frac{3x}{5x} \cdot \cos 5x = 1 \cdot 1 \cdot \frac{3}{5} \cdot 1 = \frac{3}{5}$$

1.5.4. Solution 1(d): L'Hôpital's Rule

Rewrite: $x \ln x = \frac{\ln x}{\frac{1}{x}}$

As $x \rightarrow 0^+$: $\ln x \rightarrow -\infty$ and $\frac{1}{x} \rightarrow +\infty$, so we have $\frac{-\infty}{+\infty}$.

Apply L'Hôpital:

$$\lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x \rightarrow 0^+} (-x) = 0$$

1.5.5. Solution 1(e): Limits at Infinity

Divide by x^3 :

$$\lim_{x \rightarrow \infty} \frac{3 - \frac{2}{x^2} + \frac{1}{x^3}}{5 + \frac{4}{x} - \frac{7}{x^3}} = \frac{3}{5}$$

All terms with x in denominator approach 0.

1.5.6. Solution 2(a): Chain Rule

$$f'(x) = 5(3x^2 + 2x - 1)^4 \cdot (6x + 2) = 10(3x + 1)(3x^2 + 2x - 1)^4$$

1.5.7. Solution 2(b): Product Rule

$$g'(x) = 2x \cdot e^{3x} + x^2 \cdot 3e^{3x} = xe^{3x}(2 + 3x)$$

1.5.8. Solution 2(c): Quotient Rule

$$h'(x) = \frac{2x(x^2 - 1) - (x^2 + 1)(2x)}{(x^2 - 1)^2} = \frac{2x^3 - 2x - 2x^3 - 2x}{(x^2 - 1)^2} = \frac{-4x}{(x^2 - 1)^2}$$

1.5.9. Solution 2(d): Implicit Differentiation

Differentiate: $3x^2 + 3y^2y' = 6y + 6xy'$

Solve: $3y^2y' - 6xy' = 6y - 3x^2$

$$y'(3y^2 - 6x) = 6y - 3x^2$$

$$y' = \frac{6y - 3x^2}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x}$$

1.5.10. Solution 2(e): Logarithmic Differentiation

$$\ln y = \sin x \ln x$$

$$\frac{y'}{y} = \cos x \ln x + \frac{\sin x}{x}$$

$$y' = x^{\sin x} \left(\cos x \ln x + \frac{\sin x}{x} \right)$$

1.5.11. Solution 3: Related Rates

Given: $V = \frac{4}{3}\pi r^3$

Differentiate with respect to t :

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

Solve for $\frac{dr}{dt}$:

$$\frac{dr}{dt} = \frac{1}{4\pi r^2} \frac{dV}{dt} = \frac{100}{4\pi(25)} = \frac{100}{100\pi} = \frac{1}{\pi} \text{ cm/s}$$

The radius is increasing at $\frac{1}{\pi} \approx 0.318$ cm/s when $r = 5$ cm.

1.5.12. Solution 4: Optimization

Let the rectangle have width $2x$ (from $-x$ to x) and height $y = \sqrt{R^2 - x^2}$.

Area: $A = 2x \cdot \sqrt{R^2 - x^2} = 2x(R^2 - x^2)^{\frac{1}{2}}$

Find $\frac{dA}{dx}$:

$$\begin{aligned} \frac{dA}{dx} &= 2(R^2 - x^2)^{\frac{1}{2}} + 2x \cdot \frac{1}{2}(R^2 - x^2)^{-\frac{1}{2}}(-2x) \\ &= 2\sqrt{R^2 - x^2} - \frac{2x^2}{\sqrt{R^2 - x^2}} = \frac{2(R^2 - x^2) - 2x^2}{\sqrt{R^2 - x^2}} \\ &= \frac{2R^2 - 4x^2}{\sqrt{R^2 - x^2}} \end{aligned}$$

Set equal to 0: $2R^2 - 4x^2 = 0 \rightarrow x = \frac{R}{\sqrt{2}}$

Height: $y = \sqrt{R^2 - \frac{R^2}{2}} = \sqrt{\frac{R^2}{2}} = \frac{R}{\sqrt{2}}$

Maximum area rectangle:

- Width: $2x = R\sqrt{2}$
- Height: $y = \frac{R}{\sqrt{2}}$

The optimal rectangle has width $\sqrt{2}R$ and height $\frac{R}{\sqrt{2}}$, giving maximum area $A = R^2$.

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Exercise complete. When finished, mark this task as complete to continue.